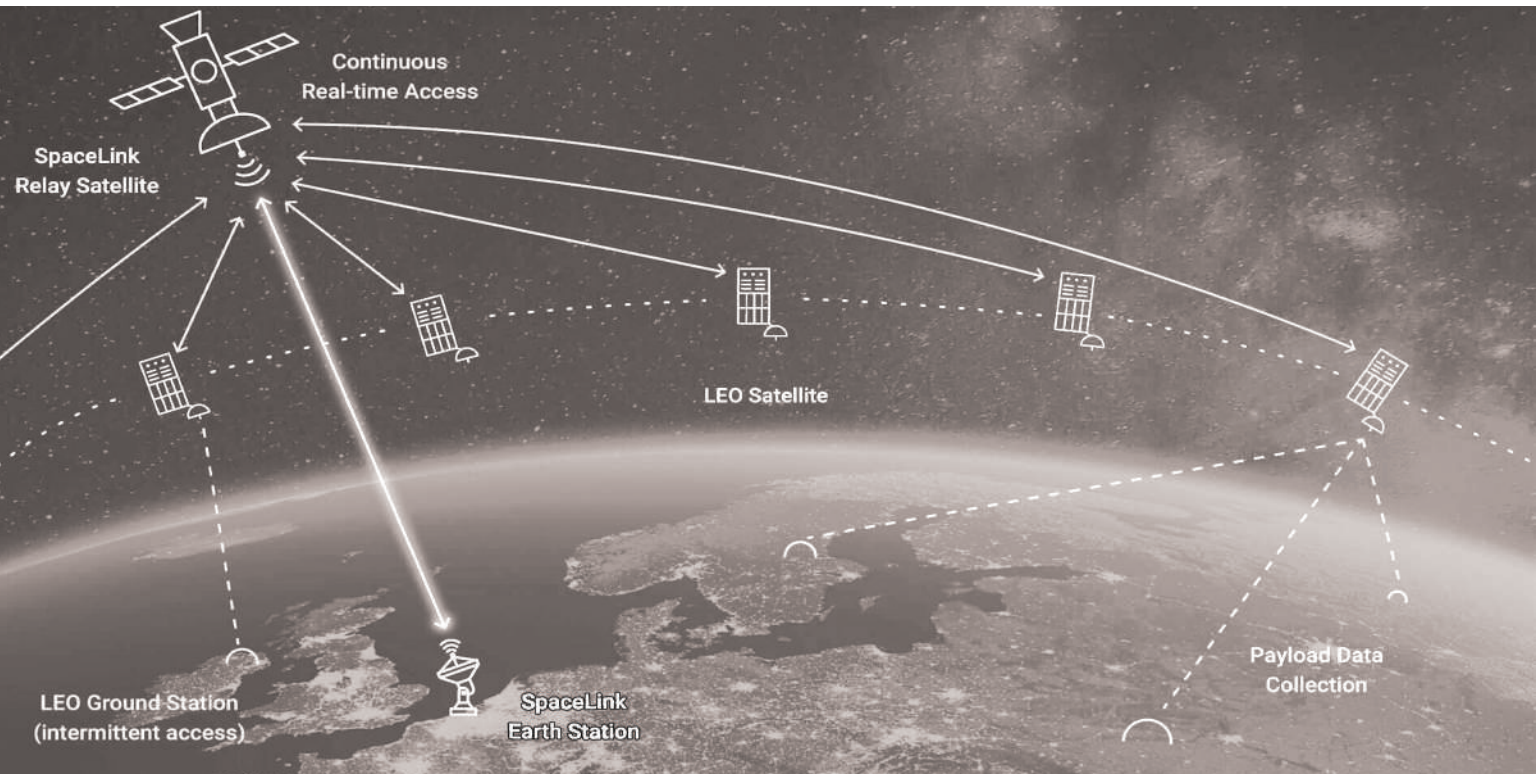


4

Linear Equations in Two Variables



Knowing the latitude, longitude, and altitude of any place in space can be used to determine its existence. Three satellites are so positioned to arrive at three equations in order to locate the position of an object at a specific location on the Earth. As a result, GPS systems follow the concept of linear equations.



What's Inside

Topic(s)

- ✓ Topic-wise Notes
- ✓ Related NCERT Qs
- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
- ✓ All Objective + Subjective Qs
- ✓ Diksha + Exemplar Qs
- ✓ Detailed Explanation

This is a **SAMPLE PDF** of our best-selling Educart Mathematics Question Bank for 2024-25. Learn for free. 😊

| TOPIC 1 |

LINEAR EQUATION

Learning Objectives

- ➔ Students will be learning the concept of linear equation in one variable and two variables.

Learning Outcomes

- ➔ Students will be able to classify the properties of a linear equation.

Real Life Application

When you attend a party or an event, you will notice decorations, the location of the celebration, meals for attendees, serving platters, and other items. Have you ever wondered how the party planner keeps it all together? In this case, the concept of a linear equation assists the party planner in creating a budget for the party or event.



A party planner always use a linear equation to calculate how much it will cost to book a venue and pay for meals per participant. A linear equation can be written to calculate the total cost, y , for any number of people in attendance, x . The party planner can use this equation to substitute any number of party guests and calculate the overall cost of the event, including food and rental charges, for her or his client.

Let's study linear equations in detail.

In earlier classes, we have learned about the linear equations in one variable (An equation which contains one variable and the highest power or degree of the variable is 1, is called a linear equation in one variable.)

$3x + 7 = 10$, $7 - 2y = 5y$, $\frac{3x}{2} + 1$, etc., are some examples of linear equations in one variable.

The linear equation in one variable is an equation which is expressed in the form of $ax + b = 0$, where a and b are two integers, and x is a variable and has only one solution.

For example: $3x + 4 = 2$ is a linear equation having a single variable in it.

Caution

→ Sometimes in question, the solution can be also be called as root, if, $ax + b = 0$

Then, its root / solution will be $x = \frac{-b}{a}$.

| TOPIC 2 |

INTRODUCTION TO THE EQUATION IN TWO VARIABLES

Learning Objectives

→ Students will be learning the general form of linear equation in two variables.

Learning Outcomes

→ Students will be able to represent any given situation in terms of a linear equation to interpret it.

Real Life Application

In every family adults work to earn for living. If their earning exceeds five lakh rupees in any year in India, then they are liable to pay the taxes. For that, they may need to find out their tax liability. Tax liability is the amount of money owed to tax authorities in the form of tax debt. It is the total amount of tax you are liable to pay to the government.



Calculations like determining the tax rebate cap and dividing earnings into proportions to determine how the income should be allocated among various



investment possibilities are part of the process of determining tax liability.

In order to arrange the revenue, spending and to determine the taxable wage, linear equations are used. This process of finding tax liability is based on doing calculations where the variables are income, the qualifying amount for the rebate, etc.

Linear equation helps the workers structure their income and expenses and get the taxable salary figure in hand. Let us now study about linear equations in two variables in brief.

An equation which can be represented in the form of $ax + by + c = 0$, where a , b , and c are real numbers and $a \neq 0$, $b \neq 0$ is called a linear equation in two variables, here in the equation x and y are variables. It is the standard form of linear equation in two variables.

$5x - y + 1 = 0$, $x - 7y = 0$, $3x - 2y + 5 = 0$ are some examples of linear equations in two variables x and y .

Important

→ $ax + by + c = 0$. Here $a \neq 0$, $b \neq 0$.

If $a = 0$, then $ax + by + c = 0$

⇒ $by + c = 0$, This is a linear equation in one variable (y). Similarly, if $b = 0$, then $b \times 0 + c = 0$

⇒ $ax + c = 0$, This is also a linear equation in one variable (x).

Now, if $a = 0$, $b = 0$ then, we get

$a \times x + c = 0$

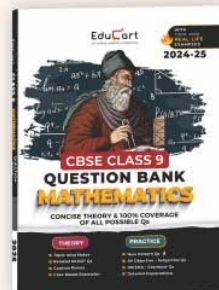
$c = 0$; No equation.

Clearly, from the above, if a or $b = 0$, then the equation $ax + by + c = 0$ become a linear equation in one variable (either in x or in y).



To continue with remaining topics of this Chapter, **click below** to purchase the complete book at a special discount.

[View Book](#)



OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. Which option shows $5y - 8x = 7(x + y) - 9$ expressed in the form of $ax + by + c = 0$?

- (a) $-x + 6y - 9 = 0$
- (b) $-x + 12y - 9 = 0$
- (c) $15x + 2y - 9 = 0$
- (d) $15x - 4y - 9 = 0$

[Diksha]

Ans.(c) $15x + 2y - 9 = 0$

Explanation: Here,

$$\begin{aligned}5y - 8x &= 7(x + y) - 9 \\ \Rightarrow 5y - 8x &= 7x + 7y - 9 \\ \Rightarrow 5y - 8x - 7x - 7y + 9 &= 0 \\ \Rightarrow -15x - 2y + 9 &= 0\end{aligned}$$

Multiply both side by -1 , we get

$$15x + 2y - 9 = 0$$

Hence, it is the standard form of the given equation.

2. Rohan bought 20 chalk boxes and duster for his school. The cost of one chalk box is ₹150 and the cost of one duster is ₹60. Rohan paid a total of ₹2000 to the shopkeeper.

Assuming that the number of chalk boxes is represented as x and the number of dusters is represented by y , which of the following equations correctly represents this situation?

- (a) $150x - 60y = 20$
- (b) $150x - 60y = 2000$
- (c) $150x + 60y = 20$
- (d) $150x + 60y = 2000$

[Diksha]

Ans.(d) $150x + 60y = 2000$

Explanation: The number of chalk boxes = x

Cost of one chalk box = ₹150

Therefore, cost of x chalk boxes = $150x$

The number of dusters = y

Cost of one duster = ₹60

Therefore, cost of y chalk boxes = $60y$



Rohan paid total ₹2000 to the shopkeeper.

Therefore, this can be represented by the equation:

$$150x + 60y = 2000$$

True and False

3. (3, 2) is a solution of the linear equation $3x - 3y = 3$.

Ans. True

Explanation: Given linear equation is

$$3x - 3y = 3.$$

Substitute the values $x = 3$ and $y = 2$ in the equation, we get

$$\Rightarrow 3(3) - 3(2) = 3$$

$$\Rightarrow 9 - 6 = 3$$

$$\Rightarrow 3 = 3$$

Yes, (3, 2) is a solution of the equation

$$3x - 3y = 3.$$

Fill in the Blanks

4. If (7, k) is a solution of the linear equation $2x - y + 6 = 0$, then the value of k is

Ans. 20

Explanation: Here, $2x - y + 6 = 0$ has a solution (7, k)

$$\text{Therefore, } 2 \times 7 - k + 6 = 0$$

$$14 - k + 6 = 0$$

$$k = 20$$

Assertion and Reason (A-R)

In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

5. Assertion (A): $9x + 15y = 6$ is a linear equation having a unique solution.

Reason (R): Linear equations in one variable has a unique solutions.

Ans.(d) Assertion (A) is false but reason (R) is true.

Explanation: The given equation, $9x + 15y = 6$ has two variables, i.e., x and y and we know that a linear equation in two variables has infinitely many solutions. Therefore, the assertion (A) is false. But, a linear equation in one variable has a unique solution.

Here, assertion (A) is false but reason (R) is true.

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

6. On her birthday, Srishti planned that this time she will celebrate her birthday in a small orphanage centre. She bought apples to give to children and adults living there. She gave 2 apples to each child and 3 apples to each adult along with Birthday cake. She distributed total 60 apples.



(A) Taking the number of children as ' x ' and the number of adults as ' y '. The above situation can be represented in linear equation in two variables as:

(a) $2x + y = 60$ (b) $2x + 3y = 60$

(c) $3x + 2y = 60$ (d) $3x + y = 60$

(B) If the number of children is 15, then find the number of adults.

(a) 10 (b) 15

(c) 20 (d) 25

(C) If the number of adults is 12, then the number of children is:

(a) 8 (b) 10

(c) 12 (d) 14

(D) If $x = -5$ and $y = 2$ is a solution of the equation $3x + 5y = b$, then value of ' b ' is:

(a) -5 (b) 5

(c) -2 (d) 2



(E) When $4y - x = 5$ is written in the form of $ax + by + c = 0$, then the value of 'a' is:

- (a) 1 (b) -1
(c) -4 (d) 4

[Delhi Gov. SQP 2022]

Ans.(A) (b) $2x + 3y = 60$

Explanation: Let the number of children be x .

Let the number of adult be y .

According to given condition $2x$ apples to each children + $3x$ apples to each adult

$$2x + 3y = 60$$

(B) (a) 10

Explanation: Given linear equation is $2x + 3y = 60$ [From (A)]

Now, put $x = 15$

$$2 \times 15 + 3y = 60$$

$$30 + 3y = 60$$

$$3y = 60 - 30$$

$$3y = 30$$

$$y = 10$$

Hence, number of adults is 10.

(C) (c) 12

Explanation: Since, number of adults is 12.

Therefore, $y = 12$

Now, put $y = 12$ in given equation $2x + 3y = 60$ we get,

$$2x + 3 \times 12 = 60$$

$$2x + 36 = 60$$

$$2x = 60 - 36$$

$$2x = 24$$

$$x = 12$$

Hence, number of children is 12.

(D) (a) -5

Explanation: Given equation is $3x + 5y = b$

On putting the values of $x = -5$ and $y = 2$ in the equation, we get

$$3(-5) + 5 \times 2 = b$$

$$-15 + 10 = b$$

$$b = -5$$

(E)(b) -1

Explanation: Given equation is $4y - x = 5$

Rewrite the equation as, $-x + 4y = 5$

On comparing the given equation with $ax + by + c = 0$

We get,

$a = -1, b = 4, c = 5,$

Hence, the value of a is -1 .

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

7. If the graph of equation $2x + ky = 10k$ intersects x-axis at point $(5, 0)$, find the value of k . [British Council 2022]

Ans. Given $2x + ky = 10k$ intersects x-axis at $(5, 0)$

$\Rightarrow (5, 0)$ lies on this line

$$2(5) + k(0) = 10k$$

$$\Rightarrow k = \frac{10}{10} = 1$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

8. Sarika distributes chocolates on the occasion of children's Day. She gives 5 chocolates to each child and 20 chocolates to adults. If number of children is represented by 'x' and total distributed chocolates as 'y'.

(A) Write it in the form of linear equation in two variables.

(B) If she distributed 145 chocolates in total, find number of children.

[Delhi Gov. QB 2022]

Ans.(A) Sonia gave 5 chocolates to each child. So, if we let number of children = x then she gave $5x$ chocolates to children. She also gave 20 chocolates to adults. So, total number of chocolates distributed = $5x + 20$.

If y = total number of chocolates distributed, then the equation is $y = 5x + 20$

(B) We have $145 = 5x + 20$



$$5x = 145 - 20 = 125$$

$$x = \frac{125}{5} = 25$$

So, there are 25 children.

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

9. In the equation shown

$-(k + 1)x + ky - 5k = 1 - 2ky$, where $k > 0$ when this equation expressed in the form $ax + by + c = 0$, gives $c = 6$. What are the values of a and b ?
[Diksha]

Ans. Given:

$$-(k + 1)x + ky - 5k = 1 - 2ky$$

$$[k > 0]$$

$$\Rightarrow -(k + 1)x + ky + 2ky - 5k - 1 = 0$$

$$-(k + 1)x + 3ky - (5k + 1) = 0$$

Multiply the equation with -1 , we get

$$\Rightarrow (k + 1)x - 3ky + (5k + 1) = 0$$

Now, compare it with $ax + by + c = 0$, we get

$$a = k + 1, \dots (i)$$

$$b = -3k \dots (ii)$$

$$c = (5k + 1), \dots (iii)$$

Since, $c = 6$

[Given]

$$\therefore 5k + 1 = 6$$

[From eq. (iii)]

$$\Rightarrow 5k = 5$$

$$\therefore k = 1$$

Now, $a = k + 1$

[From eq. (i)]

$$\Rightarrow a = 1 + 1$$

$$\therefore a = 2$$

Now, $b = -3k$

[From eq. (ii)]

$$\Rightarrow b = -3 \times 1$$

$$\therefore b = -3$$

Hence, $a = 2$, and $b = -3$

LONG ANSWER Type Questions (LA)

[4 & 5 marks]

10. If the points A(3, 5) and B(1, 4) lie on the graph of the line $ax + by = 7$, find the values of a and b .
[Delhi Gov. QB 2022]

Ans. Given there are two points

$$A = (3, 5)$$

$$B = (1, 4)$$

Also, there is a line having equation,

$$ax + by = 7$$

Now, it's given that,

These two points lie on the given line.

Therefore, they will satisfy the equation of line.

Therefore, we will get,

$$\Rightarrow 3a + 5b = 7 \quad \dots(i)$$

$$\Rightarrow a + 4b = 7 \quad \dots(ii)$$

Multiplying eqn. (ii) by 3, we get

$$\Rightarrow 3a + 12b = 21 \quad \dots(iii)$$

Subtracting eqn. (i) from (iii), we get

$$\Rightarrow 12b - 5b = 21 - 7$$

$$\Rightarrow 7b = 14$$

$$\Rightarrow b = \frac{14}{7}$$

$$\Rightarrow b = 2$$

Substituting this value in eqn. (ii), we get

$$\Rightarrow a + 4(2) = 7$$

$$\Rightarrow a + 8 = 7$$

$$\Rightarrow a = 7 - 8 \Rightarrow a = -1$$

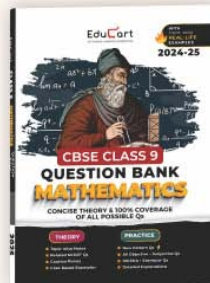
Hence, the values of $a = -1$ and $b = 2$.





For 100+ more such Questions
(including PYQ and CBSE pattern &
DIKSHA platform Q's),
please click the below button.

[View Book](#)



“ I relied on NCERT for my learning. But I practiced from Educart One Shot Books to cover new pattern Qs and PYQs. ”

SABYASACHI LASKAR

CBSE TOPPER 2023-24
(100%)



According to the CBSE topper, New pattern Qs Practice of all types and competencies are must in **Class 9** as well for each Chapter. Keeping this in mind, our Educart books cover everything important from all reference books in one place:

- ✓ Chapters divided into NCERT, PYQs and Competency
- ✓ Official Annual Exam Qs of prominent CBSE Schools (PYQs)
- ✓ Answers given separately at end for self-attempt
- ✓ Every theory point and figure summarised in Theory Maps



BUY

on Special Discount



Rita Gupta

HOD, Social Science
Reputed CBSE School

These One Shot books are the best reference books that every CBSE student should have so they can cover each Chapter in a structured manner, along with school teaching. The best part I found is the quality of answers & coverage all possible Questions

I scored 99.2% studying from Educart One Shot books. They know exactly what the students need and it really helped me do focused NCERT-driven revision and practice. A must buy book!!!



Arun Sharma

Regional Topper
CBSE 2022-23